Binomial Coefficients :

→ When n is +ve integer,

$$(a+b)^n = {}^nC_0\, a^n b^0 + {}^nC_1\, a^{n-1} b^1 + \cdots + {}^nC_n\, a^0 b^n.$$

In this expansion, the co-efficients ${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ..., ${}^nC_n$ are known as binomial co-efficients.

Res-1: ${}^nC_r = {}^nC_{n-r}$.

Res-2: ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$, $1 \le r \le n$

Res-3: ${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n = 2^n$

Ex-1: P.T $[{}^nC_0]^2 + [{}^nC_1]^2 + \cdots + [{}^nC_n]^2 = {}^{2n}C_n$
(HINT: use $(1+x)^n \cdot (1+x)^n = (1+x)^{2n}$)

Ex-2: P.T ${}^nC_1 + 6[{}^nC_2] + 6[{}^nC_3] = n^3$

Ex-3: Find n, if $C(n,3) + C(n+2,3) = P(n,3)$
[A: $n=4$].

Res-4: $(1+x)^n = \sum_{r=0}^{n} C(n,r)\, x^r = \sum_{r=0}^{n} {}^nC_r\, x^r$

Res-5 $(1+x)^{-n} = \sum_{r=0}^{n} (-1)^r\, C(n+r-1, r)\, x^r$

Res-6 $(1-x)^{-n} = \sum_{r=0}^{n} C(n+r-1, r)\, x^r$

Res-7: ${}^{n}C_{r} \cdot {}^{r}C_{k} = {}^{n}C_{k} \cdot {}^{(n-k)}C_{r-k}$,

int. $n \geq r \geq k \geq 0$ [Newton's id

Res-8: ${}^{(n+1)}C_{r} = {}^{n}C_{r} + {}^{n}C_{r-1}$. [Pascal id.]

Res-9: Vandermonde's identity:

$$C(n+m, r) = C(n,0)\,C(m,r) + C(n,1)\,C(m,r-1) + \cdots + C(n,r)\,C(m,0)$$

$$= \sum_{k=0}^{r} C(m, r-k)\,C(n,k),$$

for int. $n \geq r \geq 0$ & $m \geq r \geq 0$.